

Visvesvaraya Technological University, Belagavi – 590018



PROJECT PROPOSAL
ON
**TransGraph-PPI: A Hybrid Ensemble Framework for
Sequence-Based Protein–Protein Interaction
Prediction**

Submitted in partial fulfillment of the requirements for the degree

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE & ENGINEERING

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1 Project Title

TransGraph-PPI: A Hybrid Ensemble Framework for Sequence-Based Protein–Protein Interaction Prediction

2 Problem Statement

Proteins are essential macromolecules that drive most cellular functions, with Protein–Protein Interactions (PPIs) regulating key processes such as signal transduction, immune response, metabolism, and disease progression. Identifying PPIs is vital for understanding cellular mechanisms and discovering new therapeutic targets. However, traditional experimental methods like yeast two-hybrid assays and co-immunoprecipitation are costly, time-consuming, and not scalable to the rapidly expanding volume of protein sequence data.

Computational approaches offer scalable alternatives for sequence-based PPI prediction. Although recent deep learning models show promise, their performance often plateaus due to limitations of single-model architectures and reliance on handcrafted or structural features that may not always be available. Moreover, few methods effectively combine protein language models with graph-based learning to capture both sequence-level and network-level information.

This project proposes a hybrid ensemble framework that integrates pre-trained protein language model embeddings with a Graph Attention Network (GAT) for improved PPI prediction. A stacking-based ensemble strategy enhances robustness and generalization, while interpretability techniques provide insight into model decisions. The predicted interaction network is further analyzed to identify potential drug target proteins.

3 Objectives

The aim of the proposed project work is:

1. Design and develop a hybrid ensemble model, TransGraph-PPI that integrates ESM-2 protein language model embeddings with a Graph Attention Network (GAT) for sequence-based protein–protein interaction prediction.
2. Systematically compare the proposed hybrid model against individual standalone models (ESM-2 + MLP and GAT) and traditional machine learning approaches,

using standard performance metrics such as Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC.

3. Analyze the predicted protein interaction network using centrality-based measures and cross-reference results with drug databases to identify and prioritize potential therapeutic target proteins.

4 Software / Hardware Requirements

Software Requirements

- **Operating System:** Windows / Linux
- **Programming Language:** Python
- **Development Environment / IDE:** Jupyter Notebook, PyCharm, VS Code
- **Deep Learning Libraries:** PyTorch, HuggingFace Transformers
- **Data Processing & Scientific Libraries:** Numpy, Pandas, Scikit-learn
- **Graph Neural Network Libraries:** PyTorch Geometric
- **Explainability Libraries:** SHAP, Captum
- **Visualization Tools:** Matplotlib, Seaborn, NetworkX
- **Dataset Sources:** Public PPI databases (STRING, BioGRID, UniProt, ChEMBL)

Hardware Requirements

- **Processor:** Intel Core i5 or higher
- **RAM:** Minimum 8 GB (16 GB recommended)
- **Storage:** 20-30 GB free space
- **GPU (Optional but Recommended):** NVIDIA GPU

5 Methodology

The proposed system consists of multiple stages including data collection, preprocessing, embedding extraction, graph modeling, ensemble integration, evaluation, and network-based drug target analysis.

5.1 Data Collection

Experimentally validated human protein–protein interaction pairs will be collected from publicly available databases such as STRING and BioGRID. Protein sequence data corresponding to these proteins will be obtained from UniProt. Both positive (interacting) and negative (non-interacting) protein pairs will be included to train and evaluate the models. Drug target information will be collected from ChEMBL to validate target prioritization results.

5.2 Data Preprocessing and Feature Extraction

Protein sequences will be cleaned by removing incomplete or redundant entries. Instead of handcrafted k-mer encoding, a pre-trained protein language model (ESM-2) will be used to generate fixed-length dense embedding representations for each protein sequence. These embeddings capture contextual, evolutionary, and biochemical properties directly from raw sequence data.

For graph modeling, a protein interaction network will be constructed where proteins are represented as nodes and known interactions as edges. ESM-2 embeddings will be used as node features. Class imbalance will be addressed using weighted loss functions or appropriate sampling techniques.

5.3 Base Learner 1: Protein Language Model (ESM-2)

A pre-trained protein language model, ESM-2, will be used to generate embedding representations for each protein sequence. As a transformer-based model trained on large-scale protein data, it captures evolutionary and structural information through self-attention.

For each protein pair, the embeddings are concatenated and fed into a Multi-Layer Perceptron (MLP) classifier to predict interaction probability. This transfer learning approach enhances representation quality while avoiding training deep sequence models from scratch.

5.4 Base Learner 2: Graph Attention Network (GAT)

A Graph Attention Network (GAT) will be applied to the constructed protein interaction graph. In this representation, proteins serve as nodes initialized with ESM-2 embeddings, and interactions form the edges. The GAT model uses multi-head attention to learn importance-weighted aggregation of neighboring node features, enabling capture of higher-order topological information.

For link prediction, learned node embeddings will be combined to compute interaction probabilities between protein pairs.

5.5 Ensemble Strategy

To enhance prediction accuracy and robustness, outputs from both base models will be integrated using ensemble strategies. Soft voting will initially be implemented to combine predicted probabilities.

Additionally, a stacking-based meta-learner using Gradient Boosted Trees (XGBoost) will be trained on the probability outputs of both base models to learn an optimal combination strategy. Explainability analysis using SHAP will be applied to interpret the contributions of individual models and features to final predictions.

5.6 Performance Evaluation

The performance of the proposed system will be evaluated using standard metrics including Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC. Cross-validation techniques will be applied to ensure robustness and prevent data leakage.

The ensemble framework will be compared against individual base models (ESM-2 standalone and GAT standalone) as well as traditional machine learning baselines such as Random Forest with handcrafted features.

6 Expected Outcome

The expected outcomes of the proposed project are as follows:

- Developed a hybrid ensemble framework integrating protein language model embeddings with a Graph Attention Network for sequence-based PPI prediction.

- Achieved superior performance over individual models and traditional approaches through systematic evaluation.
- Applied SHAP and attention-based visualization to interpret interaction predictions.
- Identified potential drug targets via network analysis and implemented a scalable web-based platform for interactive PPI prediction and visualization.

References

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